ANNABELLE HERMEY

(973)-349-2714 · annabelle.hermey@gmail.com · LinkedIn

SUMMARY

Computer Science and Engineering student on the Dartmouth Women's Rowing team with a passion for problem-solving and systems. Possesses strong experience in software development, digital engineering, and control systems. Skilled at leading teams, optimizing workflows, and applying engineering principles through an internship, research, and project-based experience. Pursuing a BA in computer science (2026), a BE in computer engineering, and an MEng in computer and electrical engineering, graduating in June 2027.

EDUCATION

Dartmouth College, Hanover, NH **GPA 3.81/4.0**

Bachelor of Arts, Major in Computer Science Modified with Engineering **June 2026**

Relevant Coursework: *Object-Oriented Programming, Multivariable Calculus, Digital Engineering, Algorithms, Quantum Physics, Systems Engineering, Control Theory, Discrete & Probabilistic Systems, Software Design, Applied Computer Science, Machine Learning & Statistical Data Analysis, Full-Stack Web Development, Distributed Systems, Microprocessors*

Extracurriculars: *Women in Science Program, Alpha Xi Delta, Society of Women Engineers, Varsity Rowing*

Honors/Awards: *Citation for Academic Excellence/Merit in Machine Learning and Statistical Data Analysis, College Honor List*

Masters and Bachelor of Engineering, Electrical and Computer Engineering **June 2027**

EXPERIENCE

Engineering Intern: Falcon 9 **June 2025 – August 2025**

NASA, Marshall Space Flight Center, Commercial Crew Program

- Optimized Falcon 9 hardware certification processes for NASA's Commercial Crew Program by developing workflow automation solutions across NASA and SpaceX systems
- Automated large-scale updates of Jira filters and page structures in Confluence using Python
- Developed software programs and user interfaces to validate files and streamline data comparison and updating
- Recognized as an Honorary Crew-11 Flag Raising Honoree for exceptional performance

Teaching Assistant: Digital Engineering Lead **March 2024 – Present**

Dartmouth College, Thayer School of Engineering, Professors Geoffrey Luke and Tad Truex

- Facilitated office hours for 30+ students, providing guidance on digital engineering concepts and VHDL code
- Supported weekly laboratory projects, assisting a group of 10+ students in applying classroom concepts to the lab
- Led a team of 4-6 teaching assistants, coordinating grading, office hours, and classroom and lab support to ensure a smooth learning experience for all students; contributed to improved performance on assignments, labs, and exams
- Informally recognized by faculty and students in excellence for team-management and teaching

PROJECT-BASED EXPERIENCE

Embedded Machine Learning: Human Activity Recognition System **January 2026 – Present**

Dartmouth College, Art History Professor Chad Elias

- Developed embedded firmware in C on an STM32 microcontroller to interface with a LIS3DH accelerometer via SPI, implementing interrupt-driven sampling and deterministic real-time data acquisition
- Engineered a Bluetooth Low Energy (BLE) communication pipeline using ESP32/Arduino in C++ to stream multi-axis sensor data for real-time validation and debugging
- Trained a convolutional neural network in TensorFlow and deployed a quantized TensorFlow Lite model for low-latency, on-device inference under microcontroller memory and compute constraints
- Optimized and balanced model accuracy, RAM, and timing limitations on resource-constrained hardware

Inverted Pendulum Car Project **November 2024**

Dartmouth College, Control Theory Final Project

- Designed and implemented a proportional-integral-derivative (PID) controller to stabilize an inverted pendulum mounted on a moving car, ensuring the pendulum recovered from disturbances and maintained balance by controlling the motor to drive and reverse the car.
- Modeled system dynamics and analyzed controls using MATLAB

SKILLS

Computer: Java, Python, C, C++, MATLAB, VHDL, Assembly, HTML, JavaScript, CSS, UI/UX, Bash, shell, Git, APIs, FPGA, Arduino, STM32s, microprocessors, microcontrollers, SolidWorks, CAD, Slack, MS Office, Google Suite, Jira, Confluence

Technical: Control theory, PID Control, machine learning (AI/ML), RAG, computer vision, UART, SPI, embedded systems, soldering, actuators, sensors, data analysis, digital and analog interfacing, prototyping, iterative design,